

Design and Maintain Relational Databases

Individual Assignment

Purpose

The purpose of this assignment is to apply your SQL knowledge in a practical way to design and maintain a relational database based on an authentic case scenario. Completing this assignment, will support you later in the course when you engage in the Industry-Integrated User Case Experience, as well as, in potential future professional contexts. By completing this assignment, you will build technical and transferable skills (e.g., written communication) by:

- Executing relational database engineering tasks, including reverse engineering, schema export and integrity verification.
- Implementing MySQL queries, and communicating results to non-technical interest-holders.
- Proposing company actions or schema improvements to inform evidence-based strategy and operational optimization.
- Reflect on challenges faced, learning gained and articulating how you might apply what you have learned to future similar scenarios.

Assignment Context

A floral company is at a turning point. After years as a successful regional boutique distributor, the company has just secured a major venture capital investment to transform from a local favorite into a national floral powerhouse. To support this rapid scaling, leadership is migrating all operations from fragmented spreadsheets to a centralized, cloud-based relational database. The company CEO recently noted: *"We have plenty of data, but very little insight. In the past, we couldn't easily tell which flower families were most profitable or which customers were loyal across different cities."* As a Data Analyst on the Cloud Transition Team, your role is critical. You aren't just running code; you are building the "Source of Truth" for the company's future. Your mission is to:

- **Validate the Foundation:** Ensure the new database schema is robust and maintains data integrity.
- **Generate Intelligence:** Write SQL queries that reveal hidden patterns in customer behavior and inventory health.
- **Guide Strategy:** Translate technical results into "Plain English" insights that will help the CEO decide where to open the next shop and how to stock seasonal inventory.

Your work in this assignment directly simulates the authentic responsibility of turning raw data into a competitive business advantage.

Database

FlowerDB database contains six interrelated tables: **Flowers**, **Families**, **Shops**, **Customers**, **Orders**, and **Inventory**. These manage flower details, shop locations, customer orders, and stock levels. Refer to the ER diagram, Figure 1, or visit the database schema analysis website: <https://cind110-2026-a1.netlify.app>

Note: While FlowerDB is a fictional dataset, it is designed to mimic authentic business relationships in retail and supply chain management.

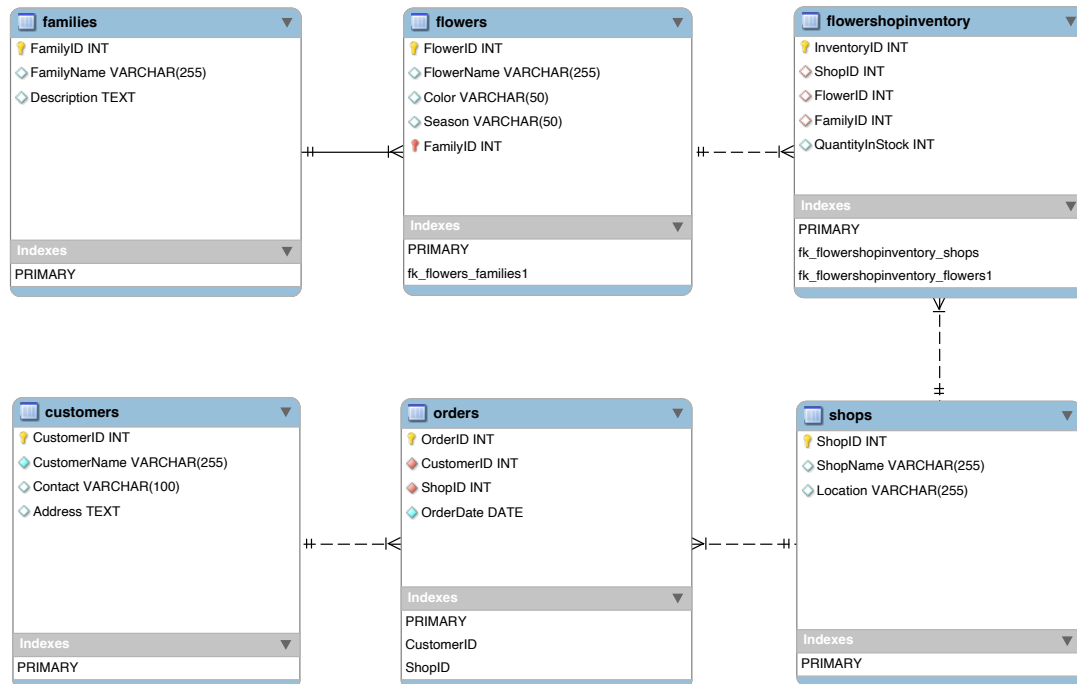


Figure 1: Entity-Relationship Diagram for FlowerDB

Academic Integrity and Use of Generative AI (Policy 60)

For **this assignment**, the limited use of Generative AI is **permitted across all project activities**, including technical troubleshooting, SQL optimization, and drafting diagnostic explanations. If you utilize Generative AI, you must adhere to the following standards:

- You must **disclose and cite** any Generative AI assistance in your Part III retrospective. Include the tool name, version, and a brief description of how it facilitated your work across the different stages of the assignment.
- You are solely responsible for the **verification and accuracy** of all SQL queries, relational models, and business strategies submitted.
- In accordance with TMU **privacy protocols**, do not paste sensitive configuration data, personal identifiers, or proprietary database schema details into external AI platforms.
- The final submission must **reflect your own knowledge and performance**; you must be able to independently explain, reproduce, and defend all technical logic and strategic recommendations.

Important Policy Restriction: Unless explicitly stated otherwise in the course syllabus or specific task instructions, the use of Generative AI is strictly prohibited for all other coursework, including **midterms, quizzes, and final examinations**. Unauthorized use in those contexts may constitute **academic misconduct** and a violation of **TMU Policy 60**.

Grading Rubric

Each of the five criteria below is graded out of **20 points**, for a total of **100 points**. To achieve an **Exceptional (16–20 points)** score in any category, your submission must meet the following performance standards:

Criteria	Exceptional Standard (16–20 Points per item)
Database Engineering	Logical model is fully accurate; all constraints (Primary/Foreign Keys) are verified with a clear technical explanation.
SQL Querying	Queries are correct, optimized, and efficient; demonstrates advanced use of joins and aggregates.
Business Interpretation	Provides actionable explanations that extract high-value patterns; connects query results to business impact.
Application of Insights	Proposes technically sound, commercially viable recommendations for management action or schema improvements based on results.
Reflection	Identifies specific technical challenges, explains the logic used to resolve them, and articulates how these skills apply to future professional roles.

Part I: Relational Engineering and Schema Validation

- **Setup:** Download the attached `FlowerDB.sql` file and load it into MySQL Workbench before beginning.
- **Task 1: Reverse Engineering**
Perform reverse engineering on `FlowerDB` to generate a logical model. Save as an `MWB` file.
Technical Resource: [MySQL Workbench Reverse Engineering Guide](#).
- **Task 2: Schema Export**
Export the complete database structure and data as a standardized SQL script.
Technical Resource: [MySQL Workbench Data Export Guide](#).
- **Task 3: Referential Integrity Verification**
Verify that primary keys, foreign keys, and constraints are enforced correctly and match the provided ER diagram (Figure 1).
 - Briefly describe how establishing **one-to-many (1:N)** and **non-identifying relationships** (dashed lines) ensures operational stability and maintains a "Source of Truth" during national scaling.
 - Provide a specific scenario (e.g., attempting to enter inventory for a non-existent `ShopID`) where a **Foreign Key constraint** would block a "phantom data" error to protect data integrity.

Part I Submission Requirements

- **Database Model:** The `.mwb` file (Task 1).
- **SQL Script:** The `.sql` export file (Task 2).
- **Primary Report:** Include the logical model screenshot and Task 3 justification in your PDF/DOCX.

Part II: SQL Analysis and Business Intelligence

For each requirement listed below, you must **Develop** optimized syntax, **Execute** (including a screenshot), **Diagnose** results in non-technical language, and **Strategize** an actionable recommendation.

*Note: To maintain professional reporting standards, please limit all query result outputs to the **top 10 rows**.*

1. **Consumer Behavior (Attribute Filtering):** Identify flowers whose color is not 'Red' and whose seasonal availability is neither 'Spring' nor 'Winter'.
Hint: Apply logical operators (NOT, IN, or <>) to isolate specific inventory attributes.
2. **Inventory Gaps (Relationship Analysis):** Identify flowers that are currently not stocked in any shop location.
Hint: Utilize an outer join or NOT EXISTS to detect missing records between tables.
3. **Client Engagement (Multi-Location Loyalty):** Generate a list of customers who have placed orders across at least two distinct shop locations.
Hint: Utilize GROUP BY and COUNT(DISTINCT ...) to evaluate multi-site engagement.
4. **Stock Volume Optimization (Nested Aggregation):** Find flower families maintaining an average stock volume of at least 140 units across the shop network.
Hint: Aggregate inventory at the (family, shop) level before calculating the cross-shop average per family.
5. **Operational Risk Monitoring (Group Constraints):** Identify shop locations where every individual inventory quantity exceeds 100 units.
Hint: Utilize the MIN() function within a HAVING clause to verify the minimum threshold for a group.
6. **Enterprise Stock Summary (Broad Aggregation):** For each flower family, display the name and the total aggregated stock quantity currently held across the entire organization.
Hint: Integrate table joins with SUM() and GROUP BY for a holistic view.
7. **Proprietary Diagnostic Query:** Formulate a business-critical question relevant to the company's expansion (e.g., "Which region has the highest concentration of loyal customers?"). Develop the SQL syntax, execute the query, and explain the strategic value of the result.

Part II Submission Requirements

- **SQL Implementation:** Raw, executable syntax for all seven queries.
- **Verification:** Screenshots of results (limited to the **top 10 rows**).
- **Business Intelligence:** Diagnostic explanations and strategic recommendations for each.

Part III: Critical Retrospective and Professional Growth

As a Data Analyst for the organization, you are required to complete a formal retrospective of your experience to drive continuous improvement. Address the following analytical prompts to demonstrate your mastery of both technical and strategic competencies.

Phase 1: Technical Retrospective (Max 100 words)

Articulate the technical execution of your tasks within the cloud migration environment:

- Summarize the specific SQL operations and diagnostic tasks you performed to validate the database.
- Identify the primary points of **technical friction** encountered during your engagement with the relational model.
- Detail the logical process and technical troubleshooting utilized to resolve these issues.

Phase 2: Strategic Alignment (Max 200 words)

Analyze the intersection of your technical output and its organizational value:

- Evaluate how the SQL queries you developed facilitate **evidence-based strategy**. Provide specific examples tied to the company's national expansion.
- What **high-value patterns** did the data reveal that would be critical for executive interest-holders?
- Identify any data gaps or schema limitations that affect the **conclusiveness** of your findings.

Phase 3: Synthesis and Transferability (Max 200 words)

Synthesize your professional development and provide future-state recommendations:

- Propose one **scalable improvement** to the FlowerDB schema or data collection methodology to enhance future predictive analytics.
- How has this engagement shaped your understanding of the Data Analyst's role in supporting operational optimization in enterprise environments?
- Identify specific technical competencies you strengthened that are directly transferable to future data-driven professional roles.

Part III Submission Requirements

To complete this section, ensure your primary PDF/DOCX report includes:

- **Analytical Narrative:** Your complete, high-quality responses for Phases 1, 2, and 3.
- **Communication Standards:** Adherence to the specified word count constraints to demonstrate concise, professional-level communication.
- **Domain Mastery:** Evidence of technical terminology and relational database concepts mastered throughout the course.

Final Submission Checklist

Final Quality Gate

- ☐ **Relational Model:** One .mwb file (Part I).
- ☐ **Database Export:** One .sql script file (Part I).
- ☐ **Comprehensive Report:** One **PDF or DOCX** file containing all screenshots, justifications (Part I), SQL diagnostics (Part II), and reflections (Part III).
- ☐ **Naming Convention:** FirstName_LastName_A1_FileType.

For technical support regarding submission portals, visit:

<https://www.torontomu.ca/courses/students/tutorials/assignments/>

End of assignment